

EEP 564 - TinyML - Assignment 2

Total Points: 100

Spring Quarter, 2026

Department of Electrical and Computer Engineering

University of Washington, Seattle, WA

Due: 11:59 pm (PST) on May 28th (Thursday), 2026 via Canvas

No Extensions

Note:

- This assignment contains both **programming questions** (marked as **[Pro]**) which are **required to write Python scripts** and **discussion questions** (marked as **[Dis]**).
- You may discuss concepts with your classmates; however, **all Python scripts and answers to discussion questions must be written and submitted independently**. Additionally, you must **declare any use of LLM / AI agents in your solution**, including where they are used and how they help with your thinking.
- Your answers to this assignment must be submitted through canvas containing the following:
 - i) **Source codes for all programming questions ([Pro])** as a **single Jupyter Notebook**, “**#_\$_EEP564_HW2_Pro.ipynb**”, where “#” and “\$” should be replaced with your first name and last name respectively. Clearly indicate which Pro problem each code block corresponds to, so your solution can be easily identified. Also, print the program’s output for each question. Make sure to include markdown cells with texts and/or in-line comments into your code for better readability.
 - ii) **Hand written and scanned or typed (e.g. word or latex) answers** to all the discussion ([Dis]) questions as **single pdf file**, namely “**#_\$_EEP564_HW2_Dis.pdf**”, where “#” and “\$” should be replaced with your first name and last name, respectively.

In this assignment, you will make use of the following dataset and files. You will find these files when you unzip the **EEP564_HW2_Files.zip**:

- **Network Anomaly Dataset**: This dataset consists of 125,973 data samples, each with 42 features corresponding to normal and attack network connections. The **network_anomaly_data.txt** file included with this homework assignment provides this dataset.
- **FirstName.LastName_EEP564_HW2.ipynb**: You will use this Python notebook file to answer Questions 2, 3, 4, and 5. When submitting your homework, **please replace FirstName.LastName with your first and last name**.
- **c_writer.py**: This script will be called in **FirstName.LastName_EEP564_HW2.ipynb** to automatically convert your tfLite model to C and create the header file (**network_model.h**) for Arduino Nano BLE deployment.
- **network_data.ino**: You will use this Arduino Sketch file to answer Question 6. You will find this file inside the **network_data** folder.

1. **[Pro]** (Data Preprocessing, 5 pts x 3 tasks = 15 pts)
Complete the following tasks in the provided Python notebook (.ipynb) file:
 - (a) Drop the 'land', 'urgent', 'numfailedlogins', 'numoutboundcmds' columns from the dataframe *data*.
 - (b) Change any label that is not named **normal** to **attack** in the {'attack'} column of the dataframe *data*.
 - (c) Use **LabelEncoder()** function from the **sklearn.preprocessing** library to convert non-numerical attributes in the 'protocoltype', 'service', 'flag', 'attack' columns of the dataframe *data* to numerical values.
2. **[Pro]** (Dimensionality Reduction for Visualization, 10 pts x 3 tasks = 30 pts)
Complete the following tasks in the provided Python notebook (.ipynb) file:
 - (a) Use **TSNE** from the **sklearn.manifold** library to visualize the data in the **test set (X_test)** in 2D. In your figure, use color **red** to mark **attack** data points and color **blue** to mark **normal** data points.
 - (b) Use **PCA** from the **sklearn.decomposition** library to visualize the data in the **test set (X_test)** in 2D. In your figure, use color **red** to mark **attack** data points and color **blue** to mark **normal** data points.
 - (c) Use **KernelPCA** from the **sklearn.decomposition** library to visualize the data in the **test set (X_test)** in 2D. Use **radial basis function (rbf)** as the kernel. In your figure, use color **red** to mark **attack** data points and color **blue** to mark **normal** data points.
3. **[Pro]** (Implementing a DNN on the dataset, 5 pts x 3 tasks = 15 pts)
Complete the following tasks in the provided Python notebook (.ipynb) file:
 - (a) Implement a **deep neural network (DNN)** on the **Network Anomaly Dataset**. Ensure to include **two neurons** and **softmax activation** in the output layer of your DNN.
 - (b) Compile and train your DNN model on the **training set (X_train)**. Denote the trained model as **base_model**.
 - (c) Evaluate the **base_model** on the **test set (X_test)** using **classification_report** and **confusion_matrix** from the **sklearn.metrics** library. **Report these numbers in your .pdf writeup file using screenshots.**
4. **[Pro]** (Implementing Quantized Model, 10 pts x 2 tasks = 20 pts)
Complete the following tasks in the provided Python notebook (.ipynb) file:
 - (a) Implement **Dynamic Range Quantization** on the **base_model**. Designate the resulting quantized ML model as **tfite_quant_model**.
 - (b) Evaluate the **tfite_quant_model** on the **test set (X_test)** using **classification_report** and **confusion_matrix** from the **sklearn.metrics** library. **Report these numbers in your .pdf writeup file using screenshots.**

5. **[Pro]** (Deploying the Quantized Model, 5 pts x 4 tasks = 20 pts)

Complete the following tasks in the provided Arduino sketch (.ino) file:

- (a) Implement code to obtain the **prediction** from the **output tensor** and determine the **predicted class label**.
- (b) Implement code to output **Sample #**, **Predicted Class**, and **Actual Class** for each sample to the serial monitor using **Serial.print** function.
- (c) Add the **network_model.h** file, generated at the end of the Python notebook, to the **network_data** folder where **network_data.ino** is kept. Upload the **network_data.ino** to the Arduino Nano BLE. **Obtain screenshots of the printed Serial Monitor outputs for five samples and report these in your .pdf writeup.**
- (d) Use the code at the end of the Python notebook to obtain **10 features and actual labels of the test set excluding the first five samples**. Add these new data at the appropriate places in the **network_data.ino**. Upload the **network_data.ino** to the Arduino Nano BLE. **Obtain screenshots of the printed Serial Monitor outputs for these ten samples and report these in your .pdf writeup.**